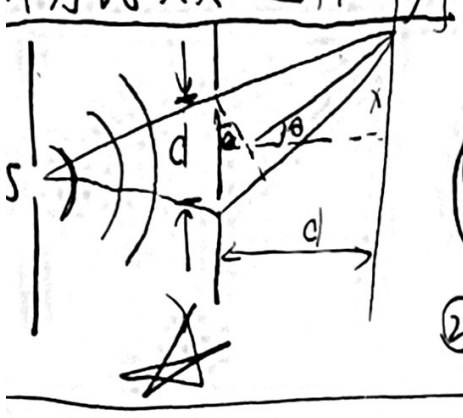
杨氏双缝干涉



① $\delta = \frac{xd}{D} = \begin{cases} \pm 2k\frac{\lambda}{2}, & \text{明} \\ \pm(2k+1)\frac{\lambda}{2}, & \text{暗} \end{cases}$ 中央为暗. $\Delta x = \frac{D}{d}\lambda$

(若放置一块 n, e 的板 $\begin{cases} \delta = \frac{xd}{D} + (n-1)e, & \text{光路} \\ \delta = (n-1)e, & \text{起始} \end{cases}$)

△近似 $d\sin\theta \approx d\tan\theta = \frac{d\cdot x}{D}$

② 条纹变化: D↑, 间距变宽; d↓, 间距变宽

薄膜干涉

① $\delta = 2e\sqrt{(n_2)^2-(n_1\sin i)^2} + [\frac{\lambda}{2}]$ △考虑半波损失 P126

⇒ 增透膜: $h = (2k+1)\frac{\lambda}{4n_c}$; 增反膜: $h = 2k\cdot\frac{\lambda}{4n_c}$

等厚干涉

1. 牛顿环

① $\delta = 2e + \frac{\lambda}{2} \pm$ (k=0,1,2,3…)

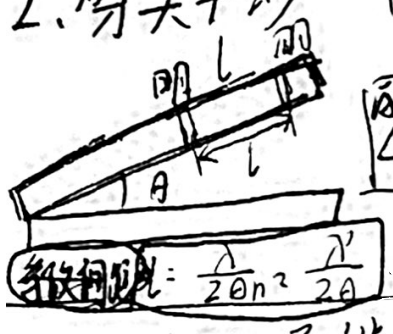
$\Rightarrow \begin{cases} \sqrt{kR\lambda}, & \text{暗环} \\ \sqrt{\frac{2k-1}{2}R\lambda}, & \text{明环} \end{cases}$ (k=1,2,3…)

(m=B) ② $d_m = \frac{r_m^2}{2R}$ (曲率半径) $\Rightarrow r = \sqrt{2eR} = \sqrt{(\delta-\frac{\lambda}{2})R}$

中心暗 ③ $\Delta r_k = \frac{1}{2}\sqrt{\frac{R\lambda}{k}}$, 中间疏, 外边密

④ 若将这玩意向上拿, 条纹如何变化! d↑(e), δ↑, 向内收缩.

2. 劈尖干涉



① $\delta = 2ne + [\frac{\lambda}{2}] = \begin{cases} 2k\cdot\frac{\lambda}{2}, & k=1,2,\dots \\ (2k+1)\frac{\lambda}{2}, & k=0,1,2 \end{cases}$ 明 / 暗

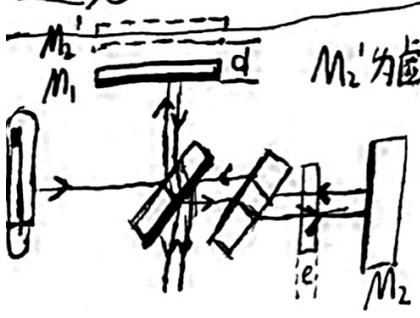
明尖端为暗纹 $\Rightarrow e = \begin{cases} \frac{2k-1}{2}\cdot\frac{\lambda}{2n}, & \text{明} \quad k=1,2,3\dots \\ k\cdot\frac{\lambda}{2n}, & \text{暗} \quad k=0,1,2,\dots \end{cases}$

$\boxed{\Delta e = \frac{\lambda}{2n}}$ (介质中)

② 条纹变化: θ↑, 向内收缩, 变密 $\Delta L = \frac{\lambda'}{2\theta}$

条纹间距: $\frac{\lambda}{2\theta n} = \frac{\lambda'}{2\theta}$

迈克耳孙干涉仪



M_2' 为虚像.

① $\delta = 2d$ (插入介质片 $\delta' = 2d + 2(n-1)e$) $+2(n-1)e = \Delta k\lambda$ $\Rightarrow e = \frac{\Delta k\cdot\lambda}{2(n-1)}$ (数条数). [Δk是干涉条纹移动数目]

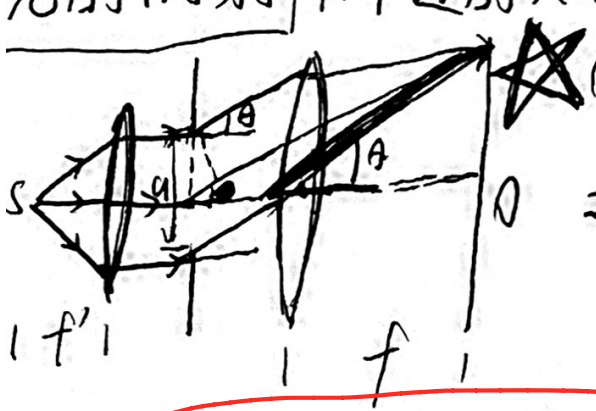
② 条件变化: Null

$2(n-1)e = \Delta k\lambda$

在镜子里多走了一倍.

光的衍射

1. 单缝的夫琅禾费衍射 中央为明纹 同侧相加, 异侧相减



① 垂直入射 $\delta = a\sin\theta$; 非垂直入射 $\delta = a(\sin\theta \pm \sin\varphi)$

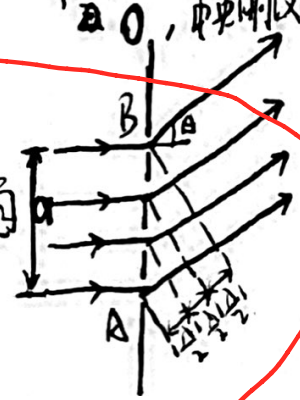
$= \begin{cases} \pm k\lambda, & \text{暗}, \text{干涉相消} \\ \pm\frac{2k+1}{2}\frac{\lambda}{2}, & \text{明}, \text{干涉加强} \end{cases}$ $\boxed{-\frac{\lambda}{2} < \theta < \frac{\lambda}{2}}$

O, 中央明纹中心

② $x_k = f\tan\theta_k \approx f\sin\theta_k = \begin{cases} 0, & \text{中央明} \\ \pm k\cdot\frac{f\lambda}{a}, & \text{暗} \\ \pm(k+1)\cdot\frac{f\lambda}{2a}, & \text{明} \end{cases}$

$\Delta x = \frac{f\lambda}{a}$ (除中央明)

※ 相邻的两个半波带对应的出射光线相干相消



③ 条件变化: 1) 缝移, 镜不动 ⇒ 衍射图样不变

2) 镜动, 缝不动 ⇒ 随镜方向而移动 镜↑, 图↑

$\begin{cases} 2k\cdot\frac{\lambda}{2}, & 2k\text{个半波带} \\ (2k+1)\cdot\frac{\lambda}{2}, & 2k+1\text{个半波带} \end{cases}$

中央明纹宽度 $\Delta x = \frac{2f\lambda}{a}$, 其他 $\Delta x = \frac{f\lambda}{a}$

艾里斑

① $\sin\theta_1 = 1.22\frac{\lambda}{d}$　② $R = 1.22\frac{f\lambda}{d}$ ⇒ 瑞利判条件

中心亮斑占总光强 84%

(1) 光学仪器的最小可分辨的视角 ⇒ 艾里斑的半角宽度：$\delta\theta = 1.22\frac{\lambda}{d}$　最小分辨角　$\delta\theta$小则$\frac{\lambda}{d}$

艾里斑半径

(2) 光学仪器分辨本领 ⇒ $\eta = \frac{1}{\delta\theta} = 0.82\frac{d}{\lambda}$　分辨本领～分辨率　分辨率 $= \frac{1}{\delta\theta} = \frac{1}{\text{角}}$

光栅衍射

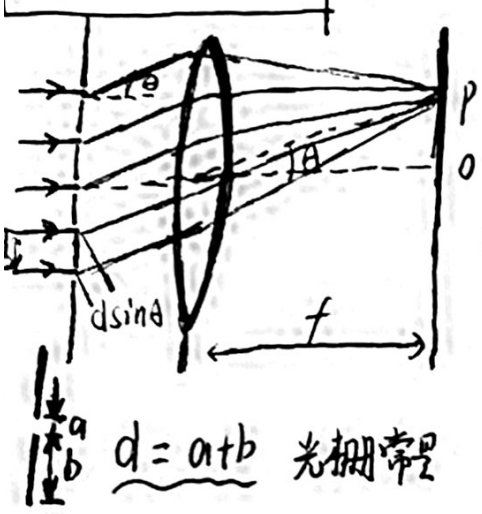
① $\delta = (a+b)\sin\theta = d\sin\theta$ ⇒ $\pm k\lambda$，明，$k = 0, 1, 2, 3, \cdots$

~~$\pm\frac{2k+1}{2}\lambda$，暗~~ 满足①，第k'次缺级.

② $k = \frac{a+b}{a}k' = \frac{d}{a}k'$ 「例：$d = 4a$，$\pm 4, \pm 8, \cdots$ 缺级」 $\neq 0$.

1个缺级，2个缺级 …

③ $k_{max} < \frac{d}{\lambda}$ 最高次



$d = a+b$ 光栅常量

同例 $\delta = d(\sin\theta + \sin\rho)$

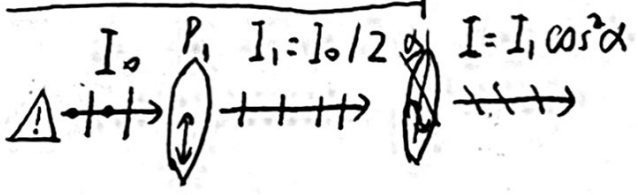
异例 $\delta = d(\sin\theta - \sin\rho)$

X射线衍射 ⇒ $\delta = 2d\sin\theta = k\lambda$ 形成亮点

马吕斯定律

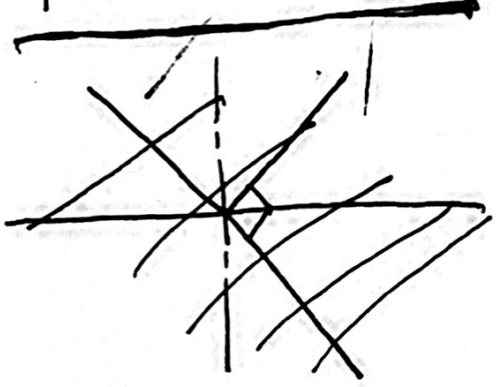
光强　入射光

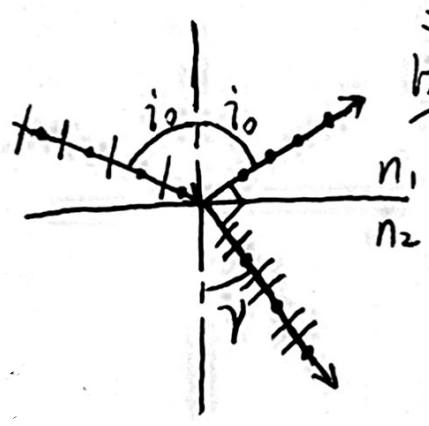
$I = I_0\cos^2\alpha$，（α为偏振片的偏振化方向与入射线偏振方向夹角）



布儒斯特定律

布儒斯特角 i_0 ⇒ $\tan i_0 = \frac{n_2}{n_1}$





⇒ 以i_0入射，反射光为s偏振光

（有 $i_0 + \gamma = 90°$）

「$n_1\sin i_0 = n_2\sin\gamma$ ⇒ $\tan i_0 = \frac{n_2}{n_1}$」

光学中两列波的相位差 $\varphi_2-\varphi_1=0$

$$\Delta\varphi=\frac{2\pi}{\lambda_1}r_1-\frac{2\pi}{\lambda_2}r_2=-\left(\frac{2\pi}{\lambda_2}r_2-\frac{2\pi}{\lambda_1}r_1\right)$$

若在不同介质中

$$\text{则}\ \Delta\varphi=\frac{2\pi}{\frac{\lambda}{n_1}}r_1-\frac{2\pi}{\frac{\lambda}{n_2}}r_2=\frac{2\pi}{\lambda}(n_1r_1-n_2r_2)$$

$$=\frac{2\pi}{\lambda}(\delta_1-\delta_2).$$